



Statistical evaluation of PM_{2.5} and dissemination of PM_{2.5}, SO₂ and NO₂ during Diwali at Chennai, India

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Abstract

The aim of this study is to evaluate the pollution caused by fireworks on the day of what is known as Diwali festival which is celebrated every year in October–November. The increase in particle pollutants such as the particulate matter 2.5 (PM_{2.5}) over a time period around the festival days is noted for a year of 2017 in the city of Chennai, South India, and this increase in pollutant levels is correlated with the increase in health issues present among public during this festival time. It was observed that PM_{2.5} varied from 30.06 ± 1.5 to 146.82 ± 7.3 $\mu\text{g}/\text{m}^3$ with an average of 68.18 ± 26.84 $\mu\text{g}/\text{m}^3$, and the mean concentration for three consecutive years (2015–2017) was lower than National Ambient Air Quality Standards. All the important pollutants such as sulphur dioxide (SO₂), nitrogen dioxide (NO₂) and PM_{2.5} have been measured during pre-Diwali, Diwali day and after Diwali at Choolaimedu (CM) and Chinmayanagar (CN) areas. Due to dense population, these areas are expected to have high levels of firework activity and are typical for the study. The gaseous pollutants SO₂ and NO₂ correlated with wind speed, wind direction, temperature and relative humidity. PM_{2.5} had good correlation with SO₂. Eye irritation was found to be the most common health issue, which affected 44.3% of the total surveyed residents during these festival days.

Keywords Fireworks · Hazardous materials · PM_{2.5} · National Ambient Air Quality Standards (NAAQS) · Diwali

1 Introduction

Diwali is one among the several religious festivals in India. It is celebrated every year during October–November, which is mostly the rainy season here. Bursting of crackers and fireworks are the major activities of this festival celebration. Fireworks are made up of chemicals such as sulphur, manganese, potassium nitrates, chlorate and perchlorates,

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carbon, sodium oxalate, aluminium, iron dust, nitrates of barium and strontium (McLain 1980). The fireworks display contains complex organic components and has trace elements raised up more than permissible values (Seidal and Birnbaum 2015; Feng et al. 2016; Kong et al. 2015; Pervez et al. 2016), which are harmful to human health (Sun et al. 2015). During Lantern festival in China, it is stated that the fireworks released the primary and secondary pollutants increasing their concentration very much (Feng et al. 2016). Here, the Diwali festival has increased such pollutant levels two to ten times. This is more than what is permitted under the Indian National Ambient Air Quality Standards (NAAQS) for levels in SO_2 , NO_2 , particulate matter-10 (PM_{10}) and total suspended particles (Kumar et al. 2016). Black carbon concentration increased by three times of its normal concentration (Babu and Moorthy 2001). During the Diwali festival in Kolkata city (Chatterjee et al. 2013) and in the 19th Commonwealth Games in Delhi, India, the air quality was measured (Bisht et al. 2013). Trace elements were polluting the atmosphere by wood combustion. But many other different types of sources are also emitting the trace metals into the environment. Fossil fuel contributes trace metals (Misch et al. 2016); automobile traffic contributes trace elements into the atmosphere (Amato et al. 2016). The firework displays are major sources of air quality deprivation in ozone (Nishanth et al. 2012).

In this study, to observe the monitoring locations in Chennai, ArcGIS pro (www.esri.com) was used. To get the temporal variations of $\text{PM}_{2.5}$ concentrations, amalgamation of MODIS (moderate resolution imaging spectro-radiometer), MISR (multi-angle imaging spectro-radiometer) and Sea WIFS (sea-viewing wide field-of-view sensor) was used. To establish the relationship between $\text{PM}_{2.5}$ and other pollutants, regression analysis of meteorological parameters were done. To indicate variation of SO_2 , NO_2 and $\text{PM}_{2.5}$, concentration levels during Diwali, scatter-plot is used.

2 Monitoring sites and methods

2.1 Monitoring sites

Chennai city is located at eastern coast of the Bay of Bengal (13.0827° N , 80.2707° E), at an altitude level 6.7 m above sea level. This city is spread out over 459 square miles and has a population of 10.265 million. Two locations, denoted as CM and CN, were selected within Chennai city for monitoring the data. Both were typical affluent residential areas who could spend for celebrating with crackers. The two are 4 km apart and have just slight variations in activities. In Fig. 1, green flags show the monitoring station of present study and black square flags show Tamil Nadu Pollution Control Board (TNPCB) monitoring stations. The house symbol flag represents the Central Pollution Control Board (CPCB) monitoring stations and red circle flag shows the area of highest PM_{10} concentration level during Diwali in Chennai.

2.2 Methodology

$\text{PM}_{2.5}$ measurement was carried with GRIMM aerosol spectrometer (Grimm 11-E). Sample air flow was automatically controlled at 1.2 L/min. Integrated and removable PTFE (polytetrafluoroethylene) filter (47 mm) was used as a gravimetric filter. The instrument was located 20 m above the ground level during festival. HTC Thermo Hygrometer (HTC-2) was used to monitor temperature and RH. The temperature range was $50\text{--}70^\circ \text{ C}$, temperature accuracy

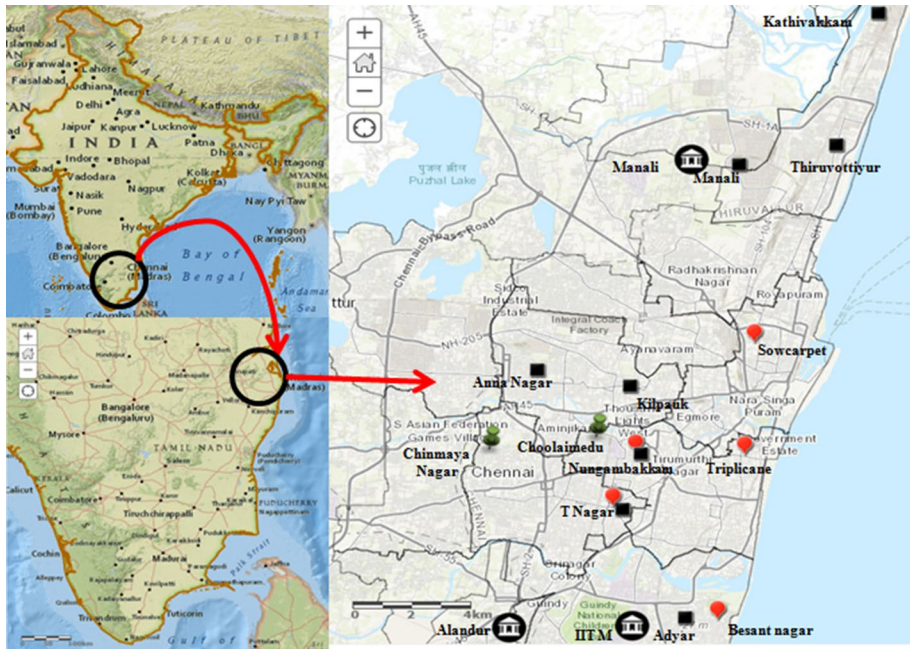


Fig. 1 Monitoring locations in Chennai using ArcGIS

and resolution were $\pm 1^\circ \text{C}$ and 0.1°C , respectively. Humidity range was 10–99%, humidity accuracy and resolution was ± 5 and 1%, respectively. The discernment of SO_2 was done by modified West and Gaeke method (West and Gaeke 1956). According to this method, SO_2 was assimilated from a known quantity of air in a solution of sodium tetra-chloro-mercurate to form di-chlorosulphitomercurate. The concentration of NO_2 in air was calculated by the method as described by Jacobs and Hochheiser (1958). Nitrogen oxide was collected in a sodium hydroxide solution to form stable sodium nitrite and was analysed spectro-photometrically. The wind speed and direction sensor (KDS-132) had wind speed range of 1.0 to 100 m/s with an accuracy and resolution of ± 1.0 and 0.01 m/s, respectively, whereas the wind direction ranged from 0° to 360° with an accuracy and resolution of ± 5 and 1° , respectively.

2.3 Calculation of a linear trend-line method

Slope

$$\alpha = \frac{n \sum (xy) - \sum x \sum y}{n \sum x^2 - (\sum x)^2} \quad (1)$$

Off-set

$$\beta = \frac{\sum y - \alpha \sum x}{n} \quad (2)$$

Trend-line

$$y = \alpha x + \beta \quad (3)$$

Slope values are combination of dependent variables (y) and independent variables (x), and it can be obtained from Eq. 1 as well as off-set values generated from β value which is denoted by Eq. 2. From Eq. 3, trend-line can be derived. Number of data points is indicated by n . Slope consists of number of data points, summation of multiple of scalar dependent variables and independent variables, multiple of summation of scalar dependent variables and summation of independent variables, summation of squared independent variables and squared value of summation of independent variables. Off-set value consists of number of data points, summation of scalar dependent variables and multiple of summation of independent variables and slope value. Trend-line is impacted by slope, off-set and independent variables.

2.4 Box plot

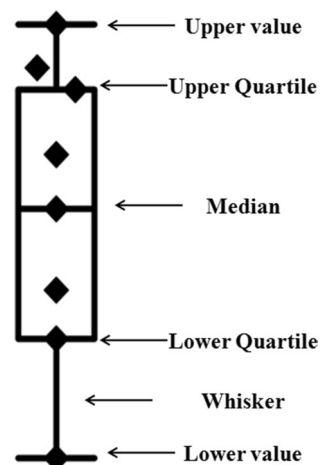
Gatherings of numerical information through their quartiles can be graphically delineating by box plot method. The change in the values of variables is used in the graph by the 25 and 75 quartiles. Box plot is used to demonstrate the shape of the dispersion, variability and its central value. The median is noted by a horizontal line inside the box. Box and data overlap type is used in present study are shown in Fig. 2 and the ambient air quality monitoring carried out during PD, DD and AD with the main festival day as DD (18th October, 2017).

3 Results and discussion

3.1 Temporal variation of $PM_{2.5}$ concentration at Manali

Figure 3 shows $PM_{2.5}$ concentration resulting from an amalgamation of MODIS, MISR and SeaWIFS. Figure 3a shows a continuous surface of concentration level of $PM_{2.5}$ in

Fig. 2 Box plot



2001–2010. From the scale, $PM_{2.5}$ concentrations level fell in the range of $21\text{--}25\text{ }\mu\text{g}/\text{m}^3$. But in Fig. 3b, $PM_{2.5}$ concentration fell in the range of $26\text{--}50\text{ }\mu\text{g}/\text{m}^3$ in 2011–2012. Annual arithmetic mean concentration was $68.18 \pm 26.84\text{ }\mu\text{g}/\text{m}^3$ for 2015–2017 shown in Fig. 4. From these, gradual increases in $PM_{2.5}$ concentration year after year can be observed.

Figure 4 shows that temporal of $PM_{2.5}$ from January 2015 to December 2017 at CPCB Manali. Manali is surrounded by industries, densely populated with a solid waste dumping yard. Monthly mean $PM_{2.5}$ concentration levels increased from 30.06 ± 1.5 to $146.82 \pm 7.34\text{ }\mu\text{g}/\text{m}^3$. Table 1 shows the $PM_{2.5}$ concentrations level during the month of September of 3 consecutive years (2015–2017) were quite below the NAAQS. In the month of April (2015- $94.32 \pm 4.71\text{ }\mu\text{g}/\text{m}^3$, 2016- $96.47 \pm 4.82\text{ }\mu\text{g}/\text{m}^3$, 2017- $85.85 \pm 4.29\text{ }\mu\text{g}/\text{m}^3$), they exceeded the NAAQS but were stable. The mean concentration in March 2015- $109.23 \pm 5.46\text{ }\mu\text{g}/\text{m}^3$, December 2016- $146.82 \pm 7.34\text{ }\mu\text{g}/\text{m}^3$ and May 2017- $87.54 \pm 4.37\text{ }\mu\text{g}/\text{m}^3$ were observed to have the highest concentration in respective years.

3.2 Relationship between $PM_{2.5}$ and other pollutants at Manali

In Fig. 5, linear regression trend-line shows better correlation between $PM_{2.5}$ and NO_2 than with SO_2 in Manali for three consecutive years (2015–2017). Mean concentrations of $PM_{2.5}$, NO_2 and SO_2 were 68.62 , 17.62 and $12.95\text{ }\mu\text{g}/\text{m}^3$, respectively. Among pollutants, $PM_{2.5}$ level exceeded the NAAQS permissible level ($60\text{ }\mu\text{g}/\text{m}^3$), whereas NO_2 and

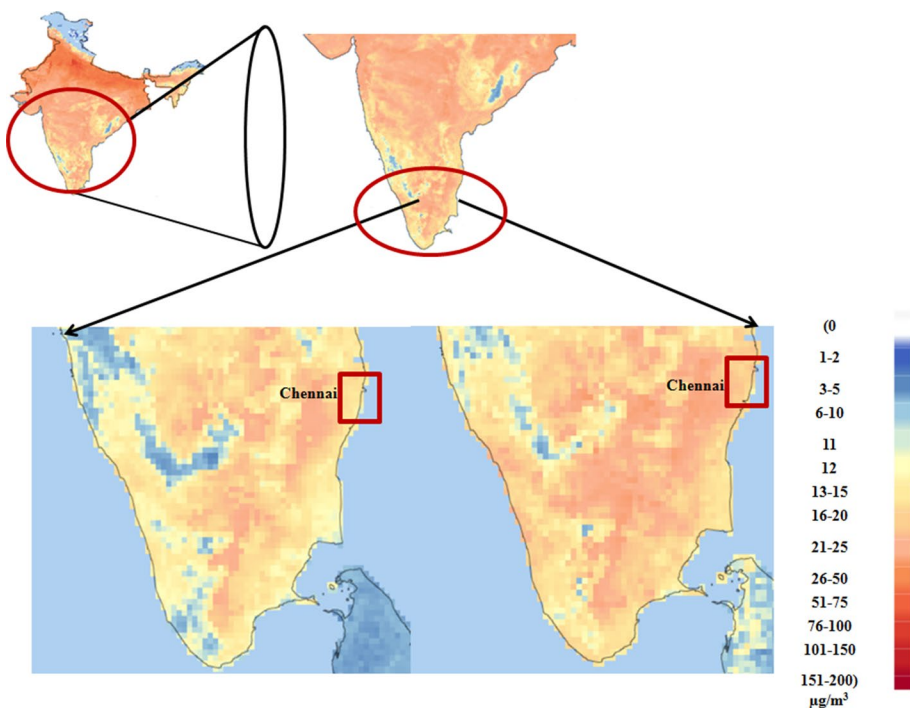
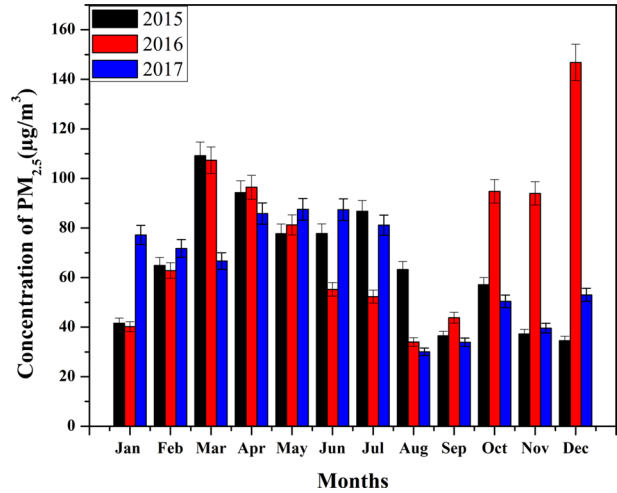


Fig. 3 a $PM_{2.5}$ Grids from MODIS, MISR and SeaWiFS, 2001–2010, b $PM_{2.5}$ Grids from MODIS, MISR and SeaWiFS, 2011–2012 (Courtesy: <https://earthdata.nasa.gov/>)

Fig. 4 Mean variation of $PM_{2.5}$ concentration during 2015, 2016 and 2017 at CPCB monitoring station Manali (Chennai)



SO_2 were below the permissible limit. NO_x concentration was below the NAAQS and the mean concentration of PM was found to be above the NAAQS (Pulikesi et al. 2006).

3.3 Regression analysis between $PM_{2.5}$ and other pollutants during Diwali

Figures 6 and 7 shows the linear regression trend-line between $PM_{2.5}$, NO_2 and SO_2 in CM and CN, respectively. At CM, NO_2 had a positive correlation with $PM_{2.5}$ but CN had a negative correlation. $PM_{2.5}$ had a better and positive correlation with SO_2 [$R^2=0.56$ (CM) and $R^2=0.82$ (CN)]. There was a positive correlation among the pollutants NO_2 and $PM_{2.5}$ (Muller et al. 2015). Due to a huge level of fireworks, SO_2 level had a positive correlation with $PM_{2.5}$ during Diwali.

3.4 Regression analysis between $PM_{2.5}$ and meteorological parameters during Diwali

Figures 6 and 7 show the linear regression trend-line of $PM_{2.5}$ with temperature, wind direction, wind speed, and relative humidity in CM and CN, respectively. In CM, $PM_{2.5}$ decreases with increase of temperature and RH. It is inversely correlated with temperature and relative humidity with coefficients of 0.04 and 0.02, respectively, at CM and $PM_{2.5}$ increases with increase of WD and WS with coefficients of 0.09 and 0.06 respectively. In CN, $PM_{2.5}$ decreases with increment in temperature, WD, WS and RH. All meteorological parameters (temperature, WD, WS and RH) were inversely correlated with factors of 0.007, 0.07, 0.03 and 0.04, respectively, at CN. It was observed that temperature, RH and wind speed had negative correlation coefficients with concentrations of $PM_{2.5}$ (Li et al. 2015, 2017). The concentration of $PM_{2.5}$ in CM was found to be higher in the direction of W and SW sectors, while the mean temperature was also found to be low ($28.78^\circ C$). WS of 1.99 ± 0.09 m/s and RH of $74 \pm 3.7\%$ were observed at the same instant. Perhaps this might be due to CM surrounded by densely populated area Thiruvalluvarpuram situated at W and SW sectors of CM. $PM_{2.5}$ concentration level in CN was observed to have a higher

Table 1 Temporal variations of PM_{2.5}

S. no.	Findings	2015 (µg/m ³)	2016 (µg/m ³)	2017 (µg/m ³)
1	Stable concentration levels (April)	94.32 ± 4.71	96.47 ± 4.82	85.85 ± 4.29
2	Below NAAQS permissible value (September)	36.5 ± 1.82	43 ± 2.18	33.92 ± 1.69
3	Highest concentration levels	109.23 ± 5.46 (March)	146.82 ± 7.34 (December)	87.54 ± 4.37 (May)

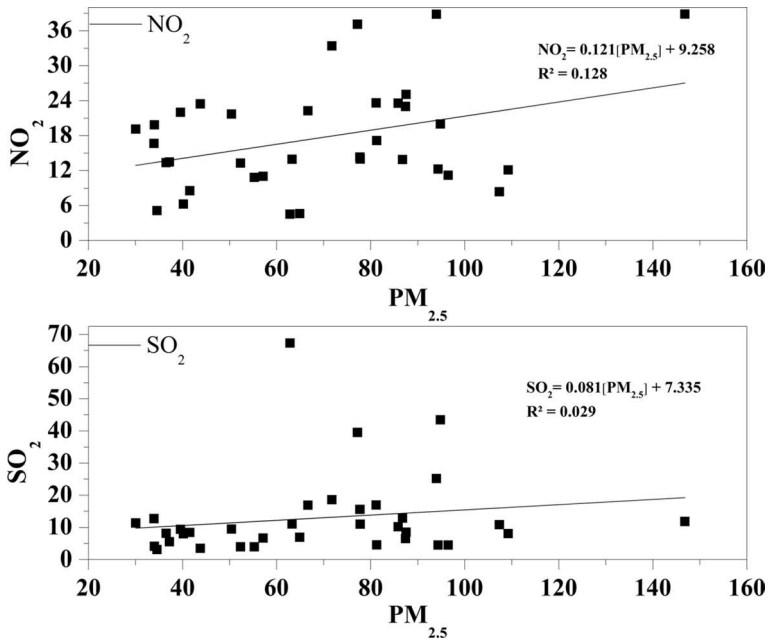


Fig. 5 Linear regression trend-line between $PM_{2.5}$ and other pollutants

wind flow in S and SSW sectors than all other wind directions sectors, but $PM_{2.5}$ concentration accumulated in SSW, SW, WSW and NW sectors.

3.5 Variation of SO_2 , NO_2 , PM_{10} and $PM_{2.5}$ concentration levels during Diwali

During Diwali festival celebration, people display fireworks in large numbers resulting in the release of pollutants and dust into the atmosphere which affects them. Measurements of SO_2 , NO_2 and $PM_{2.5}$ were carried out at CM and CN on between 16 and 21 October 2017 (festival day was on 18 October, 2017). Figure 8 shows that SO_2 , NO_2 and $PM_{2.5}$ observations which were taken DD, PD and AD at CM and CN. Figure 8a, b shows concentration of NO_2 during pre-Diwali (PD), Diwali day (DD) and after Diwali (AD) at CM and CN. On the above days, the concentration of NO_2 was within the permissible limit. Figure 8c, d shows SO_2 concentration levels during DD. Hazardous level of SO_2 was recorded between 23:00 h and 23:30 h in CM ($105.04 \pm 5.2 \mu\text{g}/\text{m}^3$), whereas in CN high levels were reached ($53.47 \pm 2.6 \mu\text{g}/\text{m}^3$) between 23:30 h and 23:59 h but stayed below NAAQS. On PD and AD, SO_2 level did not exceed NAAQS in both monitoring locations. SO_2 concentration in CM reached above NAAQS ($80 \mu\text{g}/\text{m}^3$) on DD. Figure 8e, f shows that $PM_{2.5}$ in CM reached the highest levels ($998.5 \pm 49.9 \mu\text{g}/\text{m}^3$) between 23:30 h and 23:59 h on 18 October 2017, whereas CN recorded high $PM_{2.5}$ concentration level ($947.31 \pm 47.3 \mu\text{g}/\text{m}^3$) during the same time. $PM_{2.5}$ level did not exceed NAAQS on other days. It was worthy to note that SO_2 and $PM_{2.5}$ concentration level peaked at midnight on the festival day (Chatterjee et al. 2013). Mean concentration of SO_2 , NO_2 and $PM_{2.5}$ on the festival day in CM were 33 ± 1.6 , 23.02 ± 1.1 and $215.48 \pm 10.7 \mu\text{g}/\text{m}^3$ and in CN were 16.72 ± 0.8 , 28.62 ± 1.4 and $203.29 \pm 10.1 \mu\text{g}/\text{m}^3$, respectively, as shown in Fig. 9. $PM_{2.5}$ concentration was almost

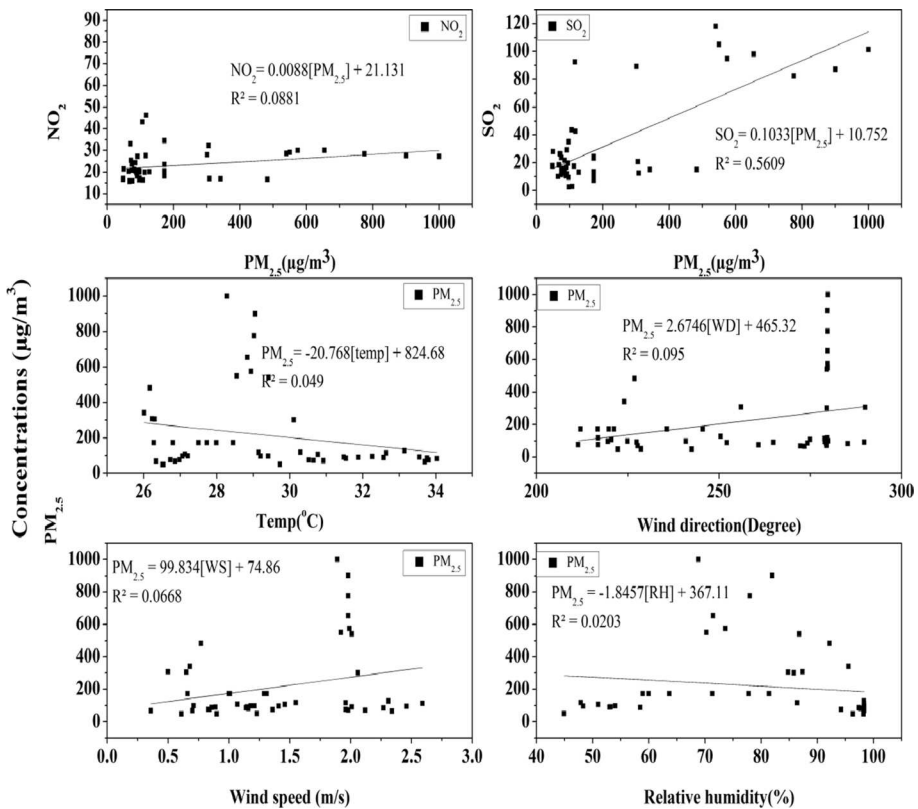


Fig. 6 Linear regression trend-line between PM_{2.5} and other pollutants and meteorological parameters at CM

three times higher than NAAQS at CN. Highest SO₂, NO₂ and PM_{2.5} concentration levels registered on DD in 2017 were 117.94 ± 5.8 , 64.83 ± 3.2 and 998.50 ± 49.9 $\mu\text{g}/\text{m}^3$, respectively. PM_{2.5} shows a sharp increase in trend (Fig. 9a, d). The concentration of NO₂ shows irregular change during study period (Fig. 9b, e). The results show that SO₂ was mentioned below limited value but after 22:30 h increased drastically (Fig. 9c, f). During DD, it can be observed from Table 2, concentration of SO₂, NO₂ and PM_{2.5} values were higher than PD and AD.

CPCB monitoring station at IIT Madras recorded dangerous levels of PM_{2.5} which was hitting 936.69 $\mu\text{g}/\text{m}^3$ between 23:00 h and 23:59 h on the festival day. Alandur recorded a high level of PM_{2.5}- 837.78 $\mu\text{g}/\text{m}^3$ between 21:00 h and 21:59 h, while Manali recorded a PM_{2.5} level of 999.9 $\mu\text{g}/\text{m}^3$ between 00:00 h and 1:59 h on next day (<http://cpcb.nic.in/>).

Figure 10a–c shows the concentration levels of SO₂ and NO₂ of previous years which were within the NAAQS limit during Diwali at five locations in Chennai city. But, PM₁₀ concentration exceeded the permissible limit during 2014 (Fig. 10a) and 2016 (Fig. 10c).

Global Asthma Network has conducted survey in India in 2014 and reported that Indian population ageing between 13 and 14 was diagnosed with current wheeze (5.8%), asthma ever (5.1%), symptoms of severe wheeze (3.4%) and symptoms of severe wheeze among current wheezers (47.8%) at 18 centres (Global Asthma Network 2014).

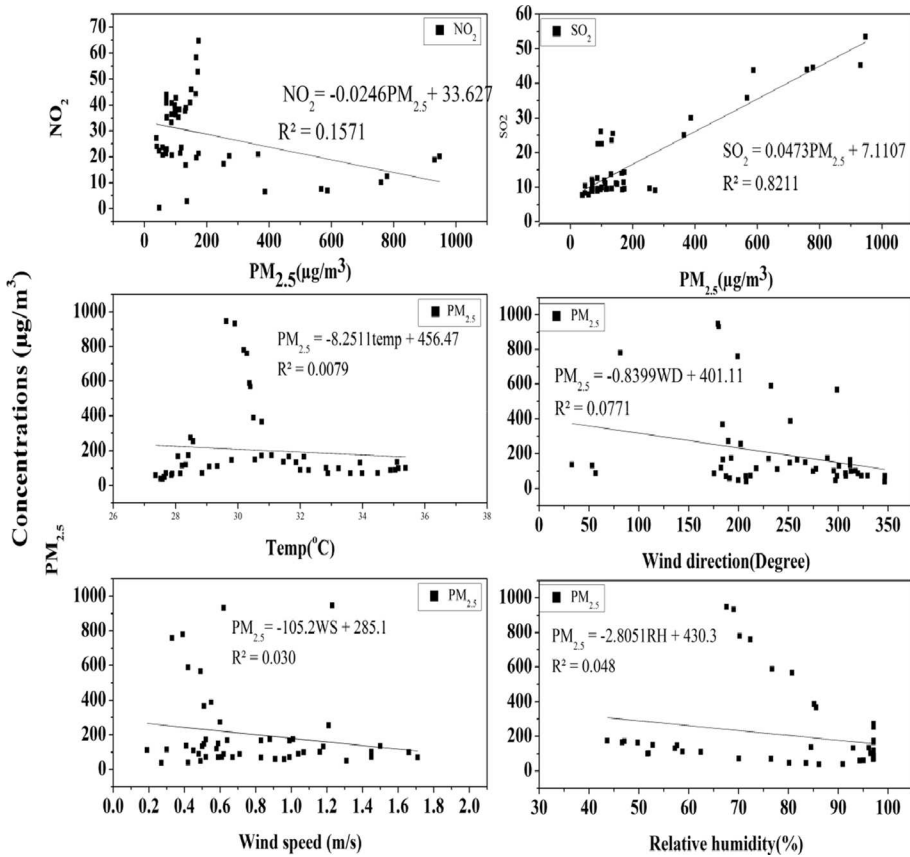


Fig. 7 Linear regression trend-line between $PM_{2.5}$ and other pollutants and meteorological parameters at CN

We conducted a health survey during Diwali on individuals residing in Chennai city, inclusive of the site we have referred to before. Few health issues were very frequent among Chennai people and its information due to environmental pollutions shown in Table 3. The acquisition of medical symptoms was acquired by field study of persons under study with medically framed questionnaire given to every person and meticulous sorting of symptoms done with the help of general physician. Eye irritation was one of the most common illnesses which accounted for 44.3% of the total survey. Cough and headache were 28.9 and 14.9%, respectively. 10% of Chennai people were affected by nausea. Vomiting was approximately 9% faced by people along with respiratory issues and throat irritation problems.

Long-term data of $PM_{2.5}$ (Grids from MODIS, MISR and SeaWiFS (2001–2012) and January 2015–December 2017) revealed that $PM_{2.5}$ concentrations varied between 25 and $146.82 \pm 7.3 \mu g/m^3$. $PM_{2.5}$ ($146.82 \pm 7.3 \mu g/m^3$) is twice than NAAQS during winter (December 2016), and it was lower ($30.06 \pm 1.5 \mu g/m^3$) during August 2017 and was very low in 2001–2010. From these observations, it is clear that $PM_{2.5}$ is increasing gradually during 2001–2017 which has caused major health issues. A significant outcome of this work is what is monitored as a mean concentration of $PM_{2.5}$ was high on AD also. From

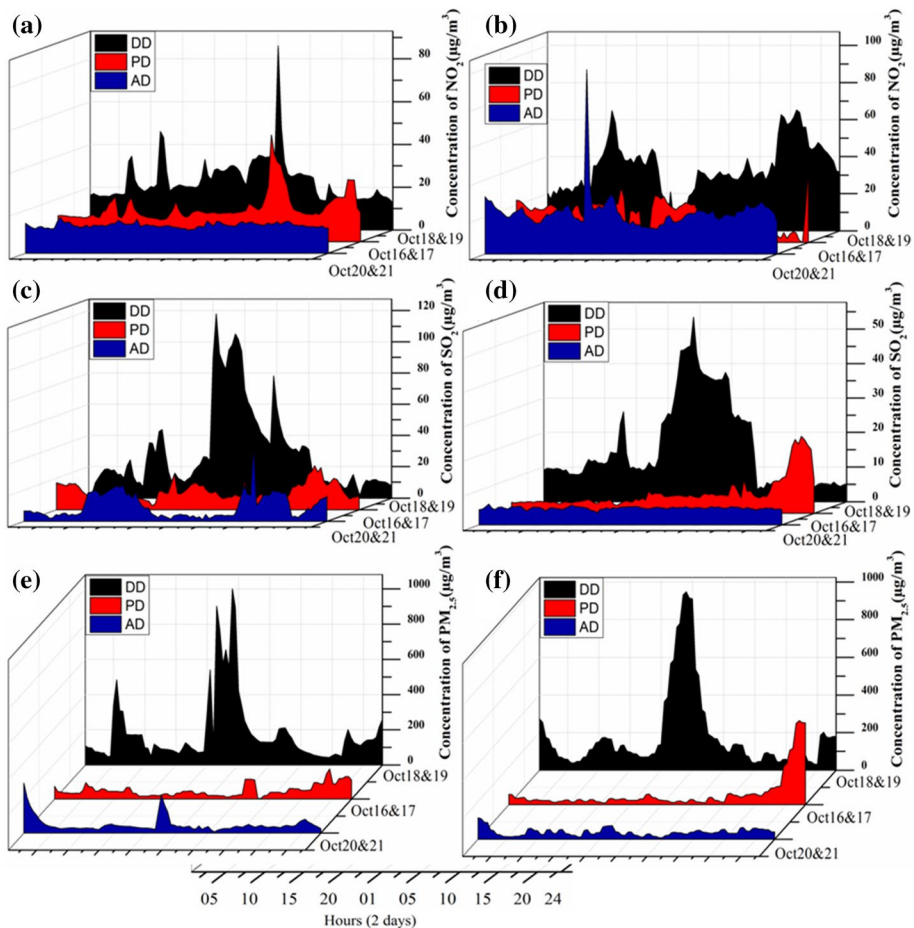


Fig. 8 Concentration of SO_2 , NO_2 , $\text{PM}_{2.5}$ during DD, PD and AD

this, it was concluded that after bursting the firecrackers, the concentration level of the pollutants settles to normal environment after a day of the festival. Instant and enduring exposure above the permissible level of NAAQS of $\text{PM}_{2.5}$ would cause environmental issues. Almost 45% of Chennai people were facing eye irritation during the festival. Eventually, this work showed that the anthropogenic pollutant contribution during the Diwali day in India has caused further decays in the air quality. Future perception is that due to restrictions now being imposed on the durations of firing crackers on Diwali day will considerably relieve the problem.

4 Conclusions

During Diwali festival, $\text{PM}_{2.5}$ emissions due to fireworks had reached almost three times higher than NAAQS though other pollutants were within permissible limits. On the festival day, regression analysis has shown that the $\text{PM}_{2.5}$ has a good correlation with SO_2 .

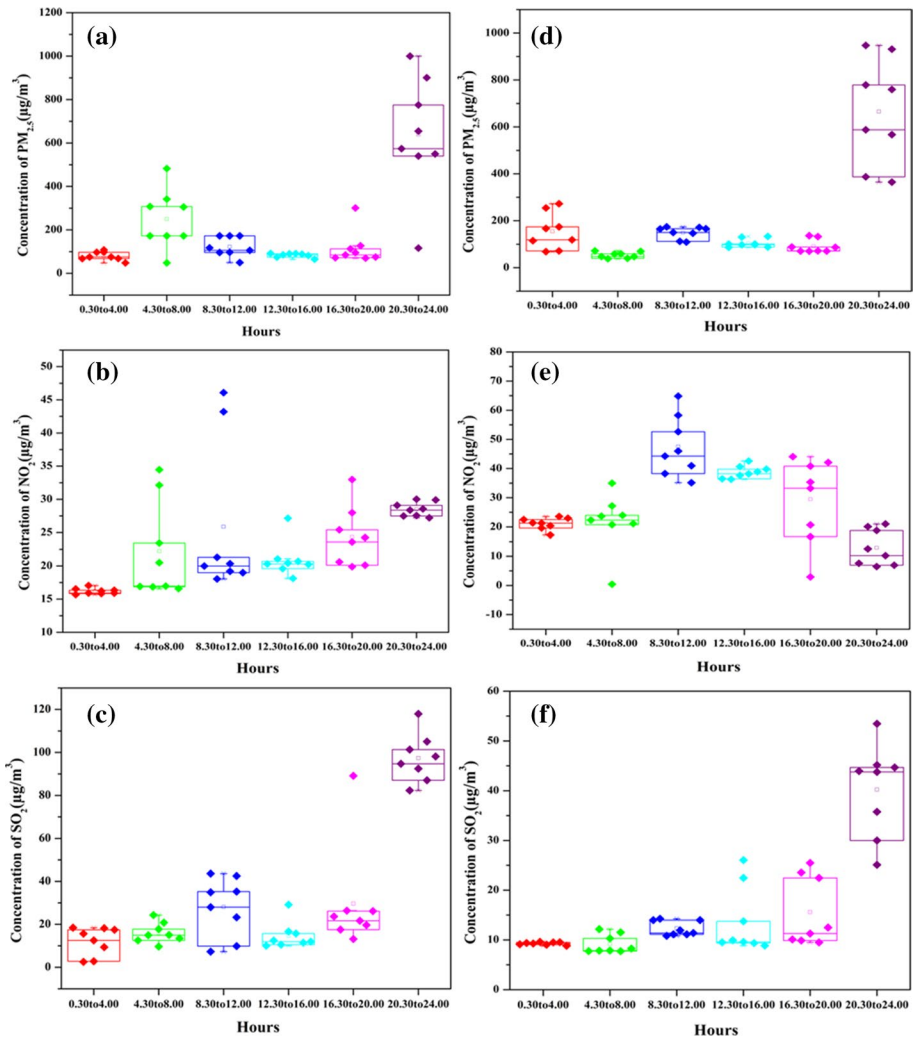


Fig. 9 Concentration of $\text{PM}_{2.5}$, NO_2 , SO_2 during festival day **a** concentration of $\text{PM}_{2.5}$ **b** concentration of NO_2 **c** concentration of SO_2 at CM **d** concentration of $\text{PM}_{2.5}$ **e** concentration of NO_2 **f** concentration of SO_2 at CN

Among all gaseous pollutants, NO_2 and SO_2 were continuously maintaining the NAAQS limits except for a few hours on the festival days. Thus, the fireworks contribute more to SO_2 which is the primary cause of $\text{PM}_{2.5}$.

Table 2 Concentration variations of pollutants on festival day and other days

Sampling locations	Pollutants	Mean concentration ($\mu\text{g}/\text{m}^3$)					
		16th Oct '17	17th Oct '17	18th Oct '17	19th Oct '17	20th Oct '17	21st Oct '17
CM	SO ₂	10.58	11.79	33.0	26.47	10.30	9.4
	NO ₂	13.23	19.12	23.02	23.41	13.12	13.13
	PM _{2.5}	38.89	59.15	215.35	153.91	58.95	36.75
CN	SO ₂	3.59	8.37	16.72	15.14	4.83	4.87
	NO ₂	18.18	10.77	28.62	38.99	24.57	20.47
	PM _{2.5}	26.86	89.85	203.29	160.18	40.02	35.0

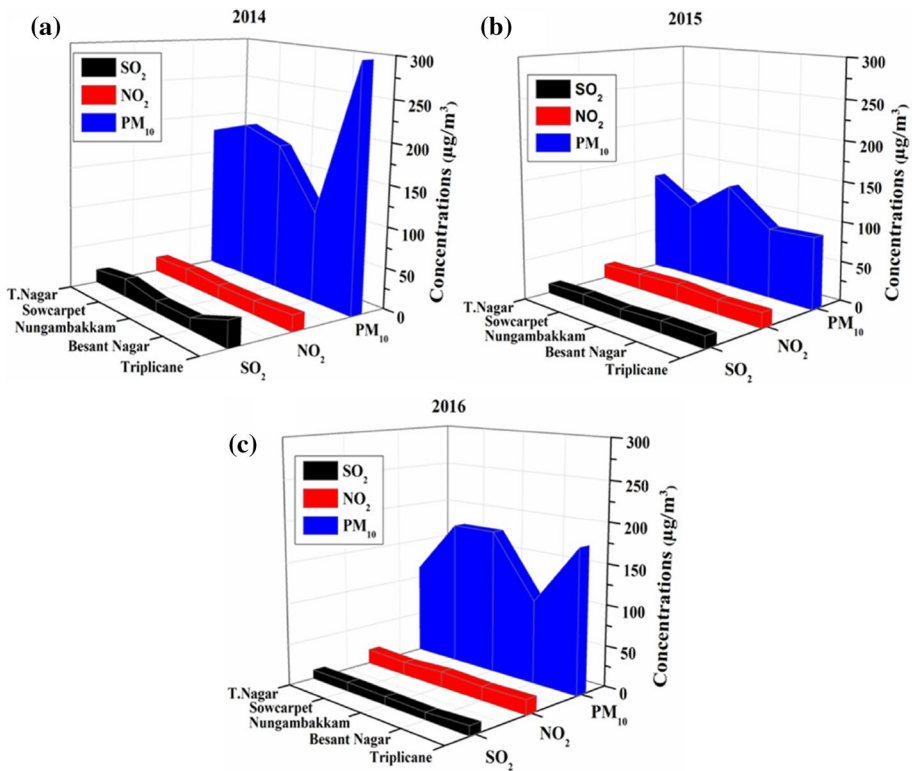

Fig. 10 SO₂, NO₂ and PM₁₀ concentration levels during Diwali

Table 3 Medical symptoms and their occurrence percentage in study area

Types of symptoms	Male (556 persons)	Female (438 persons)	Total (994 persons)	Percentage (%)
Eye irritation	249	191	440	44.3
Cough	158	129	287	28.9
Headache	55	93	148	14.9
Nausea	35	69	104	10.4
Vomiting	31	64	95	9.5
Respiratory issues	35	22	57	5.7
Throat irritation	5	2	7	0.7

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